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Programme: - B.TECH (CSE)

Assignment: - Prepare the algorithm and test case table. Then write, compile, execute , and test the C programme against the prepared use cases.

Write a menu-driven C programme using a switch-use to input a positive integer and perform one of the following operations based on the user's choice:

- 1. Find the sum of squares of its digits using a while loop.**
- 2. Find the sum of cubes of its digits using 'a' for loop**
- 3. Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop.**

Part 1: - Thorough understanding of the problem statement

Input: - 'num' (Positive integer to be processed, Variable type: 'int', Unit: unitless integer)

'choice' (Menu option selection, Variable type: 'int', Unit: unitless integer)

Output: - 'result' (The calculated sum or difference, Variable type: 'int', Unit: unitless integer)

Formula: - Digit Extraction: 'digit = n % 10' ;

Digit Removal: 'n = n / 10' ;

Sum of Squares: 'sum_sq = sum_sq + (digit * digit)' ;

Sum of Cubes: 'sum_cb = sum_cb + (digit * digit * digit)' ;

Difference: 'diff = sum_cb - sum_sq' ;

Part 2: - Writing and evaluating the algorithm

Step 1: - Start

Step 2: - Declare integer variables 'num', 'n', 'digit', 'choice', 'sum_sq'
(initialize to 0).

Step 3: - Prompt the user to enter a positive integer and store it in 'num'.

Step 4: - Validate 'num'. If 'num <= 0', display an error message and terminate the programme.

Step 5: - Display the menu options (1 for Sum of Squares, 2 for Sum of Cubes, 3 for Difference).

Step 6: - Read the user's 'choice'.

Step 7: - Initiate a 'switch case' block based on 'choice'.

Step 8: - (Case 1): Set 'n = m'. Using a 'while (n > 0)' loop: extract 'digit = n % 10', add '(digit * digit)' to 'sum_sq', and update 'n = n / 10'. Print 'sum_sq'. Break.

Step 9: - (Case 2): Using a 'for (n = num; n > 0; n / 10)' loop: extract 'digit = n % 10', add '(digit * digit * digit)' to 'sum_cb'. Print 'Sum_cb'. Break.

Step 10: - (Case 3): Set 'n = num'. Using a 'do-while' loop: extract 'digit = n % 10', update 'sum_sq' and 'sum_cb' appropriately, and update 'n = n / 10'. Continue 'while (n > 0)'. Calculate 'diff = sum_cb' and print 'diff'. Break.

Step 11: - Print an "Invalid choice" error message.

Step 12: - Stop

Part 3: - Writing and compiling the C programme

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~
Session Actions Edit View Help
GNU nano 9.0 digits.c *
#include <stdio.h>

int main() {
    int num, n, digit, choice;
    int sum_sq = 0, sum_cb = 0, diff = 0;

    // Input and Validation
    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0) {
        printf("Error: Please enter a strictly positive integer greater than zero.\n");
        return 1;
    }

    // Menu Display
    printf("\n--- Operation Menu ---\n");
    printf("1. Sum of squares of digits (using while loop)\n");
    printf("2. Sum of cubes of digits (using for loop)\n");
    printf("3. Difference between sum of cubes and sum of squares (using do-while loop)\n");
    printf("Enter your choice (1-3): ");
    scanf("%d", &choice);

    // Switch Case for Operations
    printf("\n--- Result ---\n");
    switch (choice) {
        case 1:
            n = num;
            while (n > 0) {
                digit = n % 10;
                sum_sq += (digit * digit);
                n /= 10;
            }
            printf("Number: %d, Sum of Squares: %d\n", num, sum_sq);
            break;

        case 2:
            for (n = num; n > 0; n /= 10) {
                digit = n % 10;
                sum_cb += (digit * digit * digit);
            }
            printf("Number: %d, Sum of Cubes: %d\n", num, sum_cb);
            break;

        case 3:
            n = num;
            do {
                digit = n % 10;
```

```

        sum_sq += (digit * digit);
        sum_cb += (digit * digit * digit);
        n /= 10;
    } while (n > 0);

    diff = sum_cb - sum_sq;
    printf("Number: %d, Difference: %d\n", num, diff);
    break;

    default:
        printf("Error: Invalid choice. Please select 1, 2, or 3.\n");
        return 1;
}

return 0;
}

```

Part 4: - Running and executing the algorithm on a platform (Kali Linux VM)

Step 1: - Open the terminal

Step 2: - Create the source code

Step 3: - Write and Save Code

Step 4: - Compile the programme

Step 5: - Run the Binary Executable programme

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

1 2 3 4

Session Actions Edit View Help
(kali@kali)-[~]
$ nano digits.c

(kali@kali)-[~]
$ gcc digits.c -o digits

(kali@kali)-[~]
$ ./digits
Enter a positive integer: 123

--- Operation Menu ---
1. Sum of squares of digits (using while loop)
2. Sum of cubes of digits (using for loop)
3. Difference between sum of cubes and sum of squares (using do-while loop)
Enter your choice (1-3): 1

--- Result ---
Number: 123, Sum of Squares: 14

(kali@kali)-[~]
$ ./digits
Enter a positive integer: 123

--- Operation Menu ---
1. Sum of squares of digits (using while loop)
2. Sum of cubes of digits (using for loop)
3. Difference between sum of cubes and sum of squares (using do-while loop)
Enter your choice (1-3): 2

--- Result ---
Number: 123, Sum of Cubes: 36

(kali@kali)-[~]
$ ./digits
Enter a positive integer: 123

--- Operation Menu ---
1. Sum of squares of digits (using while loop)
2. Sum of cubes of digits (using for loop)
3. Difference between sum of cubes and sum of squares (using do-while loop)
Enter your choice (1-3): 3

--- Result ---
Number: 123, Difference: 22

(kali@kali)-[~]
$ ./digits
Enter a positive integer: 45

--- Operation Menu ---
1. Sum of squares of digits (using while loop)
2. Sum of cubes of digits (using for loop)
```

```

(kali㉿kali)-[~]
$ ./digits
Enter a positive integer: 45

--- Operation Menu ---
1. Sum of squares of digits (using while loop)
2. Sum of cubes of digits (using for loop)
3. Difference between sum of cubes and sum of squares (using do-while loop)
Enter your choice (1-3): 1

--- Result ---
Number: 45, Sum of Squares: 41

(kali㉿kali)-[~]
$ ./digits
Enter a positive integer: -10

```

```

(kali㉿kali)-[~]
$ ./digits
Enter a positive integer: -10
Error: Please enter a strictly positive integer greater than zero.

```

Step 5: - Standard test table

TEST CASE	INPUT (num, choice)	EXPECTED OUTPUT	ACTUAL OUTPUT	RESULT (PASS/FAIL)
1	num = 123 choice = 1	Sum of Squares: 14	Sum of Squares: 14	PASS
2	num = 123 choice = 2	Sum of Cubes: 36	Sum of Cubes: 36	PASS
3	num = 123 choice = 3	Difference: 22	Difference: 22	PASS
4	num = 45 choice = 1	Sum of Squares: 41	Sum of Squares: 41	PASS
5	num = -10 choice = 2	Error: positive integer	Error: positive integer	PASS

Part 6:- Evaluation

The algorithm executes with a time complexity of $O(d)$, where d is the total number of digits in the input integer (equivalent to $O(\log_{10}(\text{num}))$).